

PROJECT DOCUMENTATION

Project: Solving Inventory Inefficiencies Using SQL

1. PROJECT OVERVIEW :

This project aims to simulate the responsibilities of a data analyst working in a retail environment. My goal was to build an SQL-driven inventory monitoring system which can:

- Track stock levels
- Detect low inventory conditions
- Estimate reorder points using historical sales
- Analyze inventory turnover
- Support smarter, data-backed decisions using KPIs

I started by designing a clean database schema, moving on to writing scalable SQL scripts, and preparing actionable outputs that were further visualized in a Power BI dashboard.

(NOTE : pdf shows only limited entries of various KPIs , while the .pbix file consists of all entries .)

2. SCHEMA USED :

I structured the database into three core tables and one temporary staging table:

Table Name	Description
raw_inventory	Temporary staging table for the raw CSV import. Used for transformation.
Stores	Contains store identifiers and their respective region information.
Products	Stores product IDs, pricing, and discounts.
Inventory_Records	Central fact table containing daily inventory data, sales, demand forecasts, weather, and other dimensions.



3. TECHNOLOGIES USED IN PROJECT :

- MySQL Workbench for schema creation and query writing
- Jupyter Notebook for importing raw_inventory table
- Power BI Desktop for visualizing KPIs and presenting the final dashboard

4. SQL WORKFLOW BREAKDOWN :

The SQL script is structured in well-defined blocks to promote clarity and ease of execution:

Step-1 : Table Creation

All main tables were created using normalized structures. Data types were chosen to balance efficiency and precision.

Step-2: Data Insertion from Raw File

The raw_inventory CSV file was imported, and data inserted into cleaned tables using transformations (store ID formatting, date conversions, etc.).

Step-3: Key Analytical Queries

Queries included low inventory detection, reorder point estimation, turnover analysis, sales decline detection, and days of inventory left. All were optimized for performance.

Step-4: Index Creation

Indexes were added to improve performance on frequently queried columns:

- store_id, product_id, date in Inventory_Records , Inventory levels and region of stores.